

Automate FW testing with MDBCore scripting, part of the MPLAB[®]X ecosystem (module2: compile without IDE)

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Nov 2023

Agenda

1: what is the *MPLABX* ecosystem *MDBCore*, *SDK* -> how to use for automated FW testing

2: compile without IDE

3: Automated *FW-testing* with different *MDBCore* use-models

- a. simple *mdb* -> DOS/bash-script, easy to use but has limits
- b. more advanced *mdbcs* -> *.jar module needs scripting-language with Java-interface, more complex but solves *mdb* limits and opens door into Java-universe
- c. full, but most complex usage of *MDBCore* with Java -> SDK

4: summary

M2: Steps from *.c/.h to *.elf

Three steps to compile your FW into an *.elf:

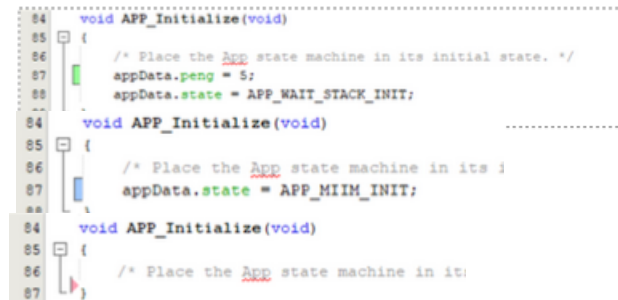
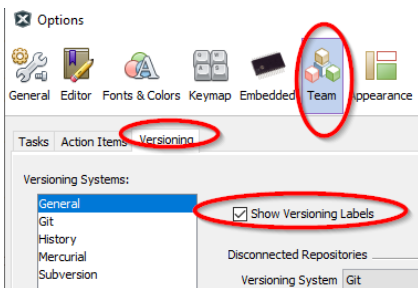
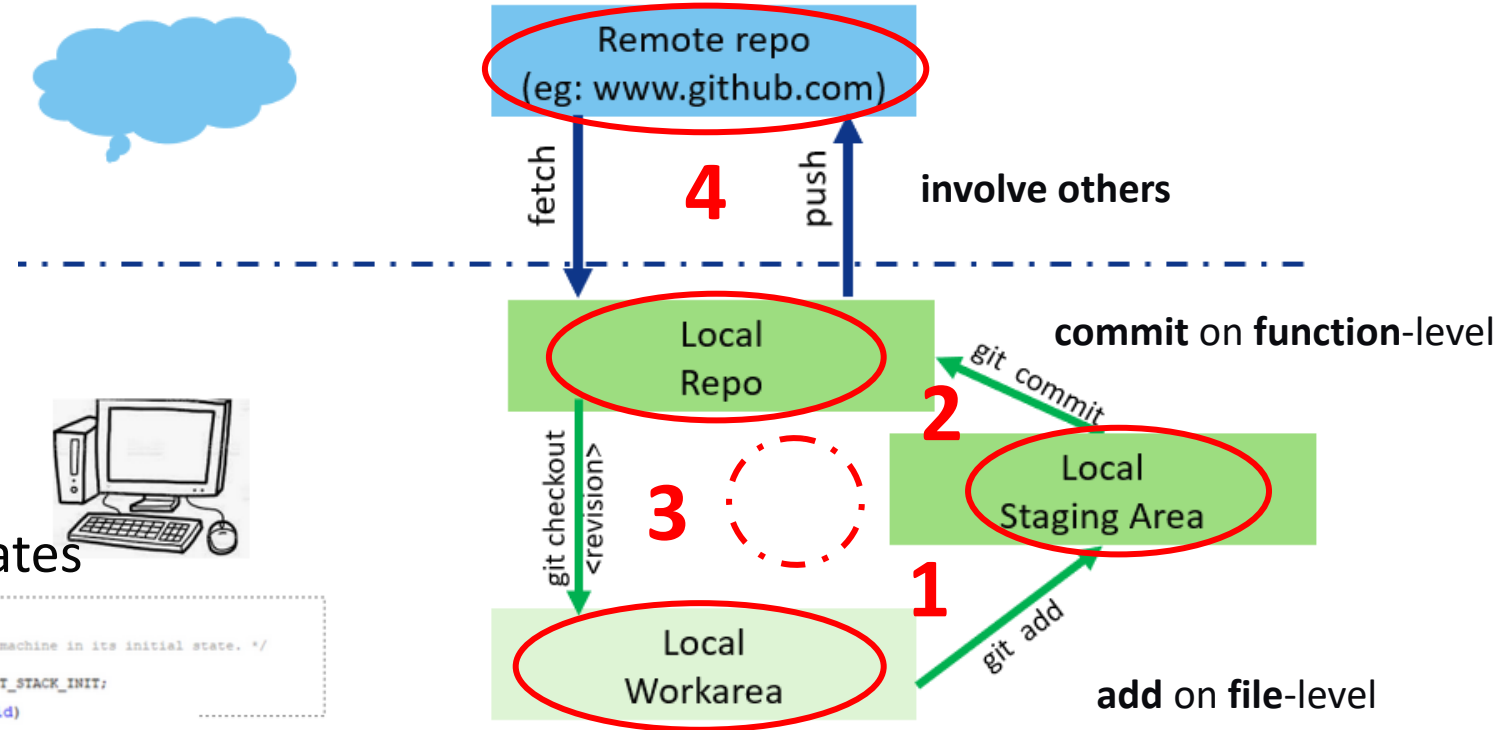
1. First you need to somehow retrieve a certain revision of your FW (->In: tag => Out: configuration.xml & <revA>*.c/.h)
2. Create Makefiles (->In: configuration.xml => Out: nbproject/Makefile*)
3. Compile (->In: nbproject/Makefile* => Out: dist/.../*.elf)

M2: checkout sources from SCM

FW revisions with SourceControlManagement/SCM

- Different tools like **git**, **subversion**,... are available to hold various FW-revision
- Common to all are features like tags, branches to define what a FW-revision-A contains
- SCMs typically have 3 major components, **git** even has 4
- **git** workflow
- git **tags** 'communication ok', 'uartRX working'
- git **branches** 'uartRX polled' vs 'uartRX ISR-driven'
- MPLABX color markings for git-states

just simple intro, teaching
git beyond scope of class



M2: parts of a MPLABX project

a MPLAB^(R) >

<project-pat

| -<projectl

| | -Make

| | -mcc_

| | -nbpro

| | | -p

| | | -c

| -src\ -> yc

Summary of files+directories to checkin:

- project.X\
 - Makefile
 - *.mc3
- nbproject\
 - project.xml
 - configuration.xml
- src\

```
200 </logicalFolder>
201 <logicalFolder name="f2" displayName="clk" projectFiles="true">
202   <itemPath>../src/config/sam_v71_xult/peripheral/clk/plib_clk.c</itemPath>
203 </logicalFolder>
204 <logicalFolder name="f3" displayName="dacc" projectFiles="true">
205   <itemPath>../src/config/sam_v71_xult/peripheral/dacc/plib_dacc.c</itemPath>
206 </logicalFolder>
```

M2: Steps from *.c/.h to *.elf

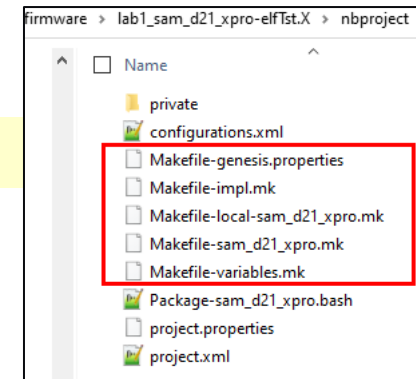
Steps to compile your FW into an *.elf consists on these three steps:

1. First you need to somehow retrieve a certain revision of your FW (->In: tag => Out: configuration.xml & <revA>*.c/.h)
input step1: <revisionA>

-> <extract revA - eg: git checkout revA>

<revisionA>*.c/.h + configuration.xml

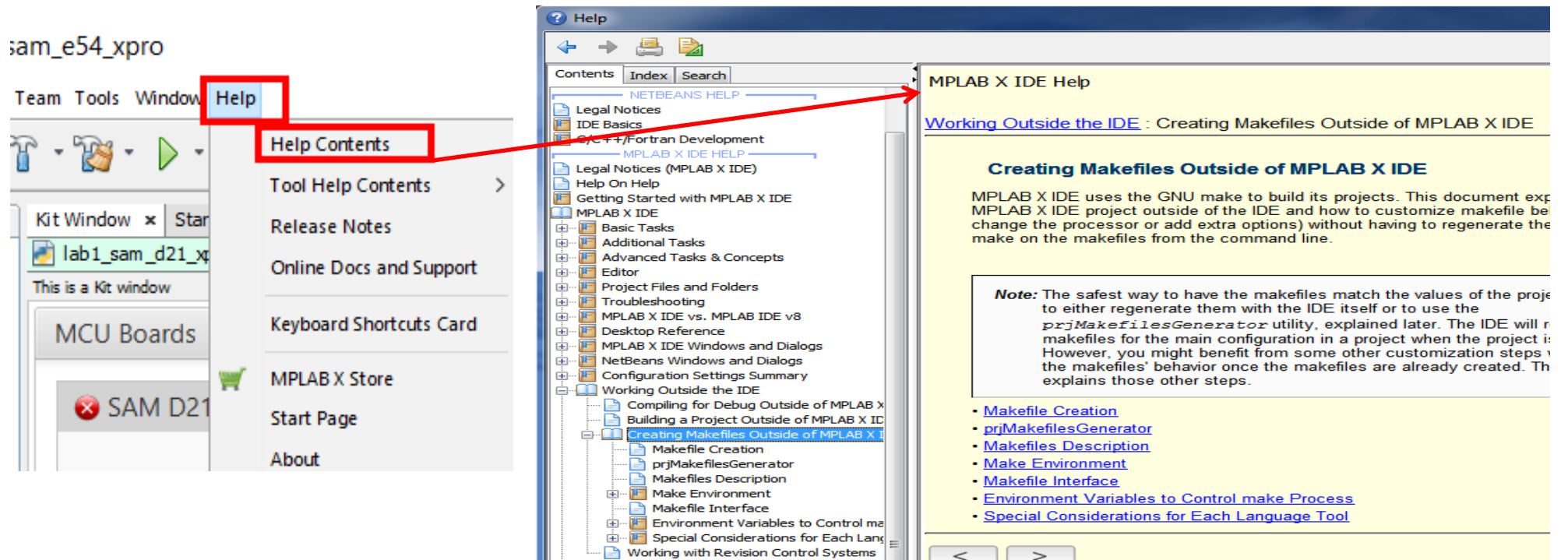
2. Create Makefiles (->In: configuration.xml => Out: nbproject/Makefile*)



3. Compile (->In: nbproject/Makefile* => Out: dist/.../*.elf)

M2: Help documentation

- MDBCore provides a Java-module with tool [*prjMakefilesGenerator.jar*](#) to update/create the Makefiles based on the info found in [*configuration.xml*](#)
- Provided as [*prjMakefilesGenerator.bat*](#) calling the MDBCore-module
- Where is it? => `<mplabx-v>\mplab_platform\bin\`
- more details on the next slide and in MPLAB help 'Working Outside the IDE'



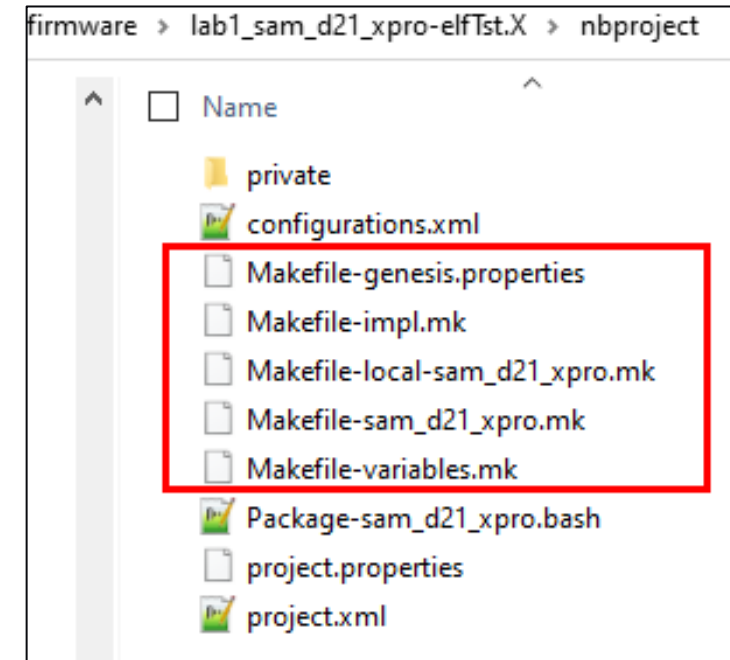
M2: creating Makefiles

Using the prjMakefilesGenerator.bat

- syntax

```
(DOS) > prjMakefilesGenerator.bat \  
        C:\myprj\prj2.X \  
        "C:\my prj\prj1.X@default@sam_d21_xpro"
```

!! DoubleQuotes to handle with spaces !!



- Which Makefiles are updated?
 1. In prj2.X **all** the MPLAB®X configurations
 2. In prj1.X **only** MPLAB®X configurations named *default* and *sam_d21_xpro*

M2: Steps from *.c/.h to *.elf

Steps to compile your FW into an *.elf consists on these three steps:

1. First you need to somehow retrieve a certain revision of your FW (->In: tag => Out: configuration.xml & <revA>*.c/.h)
input step1: <revisionA>

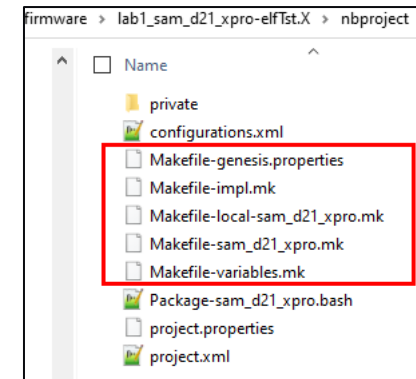
-> <extract revA - eg: git checkout revA>

<revisionA>*.c/.h + configuration.xml

2. Create Makefiles (->In: configuration.xml => Out: nbproject/Makefile*)
input step2: configuration.xml

->prjMakefilesGenerator.bat

output step2: nbproject\Makefile-*



3. Compile (->In: nbproject/Makefile* => Out: dist/.../*.elf)

M2: compile with Make

Finally, call *make* with different options, depending on intention:

- Re-compile with *make* to re-create *.elf
- *make* comes with MPLAB® X installation -> *<mplabx>\gnuBins\GnuWin32\bin*
- *make* calls central *Makefile* *inside* the project directory *prj.X*
- Different make calls for different results

#- project has *only ONE* configuration

```
DOS> cd <path-2-prj>\prj.X
```

#- change *into* project directory

```
DOS> make clean
```

#- remove all intermediate+final objects

```
DOS> make TYPE_IMAGE=DEBUG_RUN
```

#- create *debug-code* / *.elf

#- project has *only MULTIPLE* configurations

```
DOS> make CONF=myConfig
```

#- *production-code* / *.hex for *myConfig*

- new dbg-*.elf ::

```
prj.X\dist\default\ [debug,production]\<project>.X. [debug,production].elf
```

M2: cleanup and preparation

- Cleanup temp files&directories

```
path\  
| -project.X\  
|   | -Makefile  
|   | -myconfig.mc3  
|   | -nbproject\  
|   |   | -project.xml  
|   |   | -configuration.xml  
| -src\  
|
```

- extend OS-search-path

- `<mplabx-v>\mplab_platform\bin\` -> `prjMakefilesGenerator.bat`
- `<mplabx-v>\gnuBins\GnuWin32\bin\` -> `make.exe`

- checks to verify

```
C:\Windows\System32\cmd.exe  
  
C:\mchp\Masters\2021_eMasters-MU\24024DEV11_eMasters\Firmware\docu>set|grep Path  
Path=C:\Program Files\Microchip\MPLABX\v5.45\gnuBins\GnuWin32\bin;C:\Program Files\Microchip\MPLABX\v5.45\mplab_platform\bin;C:\Program Files\Microchip\xc32\v2.5  
0\bin;C:\Program Files\Microchip\xc16\v1.61\bin;C:\Program Files\Microchip\xc8\v2.31\bin;C:\Program Files\Microchip\MPLABX\v5.45\sys\java\zulu8.36.0.1-ca-fx-jdk8  
.0.202-win_x64\jre\bin;C:\PROGRA~2\Groovy\Groovy-2.5.8\bin;C:\Program Files\git\v2.25.1\Git\bin;C:\WINDOWS\system32;C:\WINDOWS;C:\WINDOWS\System32\Wbem;C:\WINDOW  
S\System32\WindowsPowerShell\v1.0\;C:\WINDOWS\System32\OpenSSH\;C:\Program Files (x86)\Webex\Webex\Applications;C:\Program Files\TortoiseSVN\bin;C:\Program Files  
(x86)\Sennheiser\SoftphoneSDK\;C:\Program Files (x86)\Common Files\Oracle\Java\javapath;C:\opt\gnu_tools_arm\9-2020-q2-update\bin;C:\opt\gnu_tools_arm\9-2020-q2  
-update\arm-none-eabi\bin;C:\opt\msys64;C:\opt\msys64\usr\bin;C:\Program Files (x86)\Webex\Plugins;C:\Program Files\TortoiseGit\bin;%PYTHON38_P%;%PYTHON27_P%;C:\  
mchp\00GnuToolsA11\v14-04-03;C:\Python\v3.8.2\Scripts;C:\Python\v3.8.2\;C:\Users\m16422\AppData\Local\Programs\Python\Python37\Scripts;C:\Users\m16422\AppData\  
Local\Programs\Python\Python37\;C:\Users\m16422\AppData\Local\Microsoft\WindowsApps;C:\Program Files\git\v2.21.0\Git\cmd;C:\Users\m16422\AppData\Local\Gpg4win\..  
\GnuPG\bin;  
PSModulePath=C:\Program Files\WindowsPowerShell\Modules;C:\WINDOWS\system32\WindowsPowerShell\v1.0\Modules  
  
C:\mchp\Masters\2021_eMasters-MU\24024DEV11_eMasters\Firmware\docu>  
C:\mchp\Masters\2021_eMasters-MU\24024DEV11_eMasters\Firmware\docu>  
C:\mchp\Masters\2021_eMasters-MU\24024DEV11_eMasters\Firmware\docu>where make  
C:\Program Files\Microchip\MPLABX\v5.45\gnuBins\GnuWin32\bin\make.exe  
C:\opt\msys64\usr\bin\make.exe  
C:\mchp\00GnuToolsA11\v14-04-03\make.exe  
  
C:\mchp\Masters\2021_eMasters-MU\24024DEV11_eMasters\Firmware\docu>  
C:\mchp\Masters\2021_eMasters-MU\24024DEV11_eMasters\Firmware\docu>  
C:\mchp\Masters\2021_eMasters-MU\24024DEV11_eMasters\Firmware\docu>where prjMakefilesGenerator.bat  
C:\Program Files\Microchip\MPLABX\v5.45\mplab_platform\bin\prjMakefilesGenerator.bat
```

M2: Compile without IDE - Knowledge Check

Summary and knowledge check module2: compile without IDE

- Where are the different FW-revision stored?
=> In a SCM-system like git, subversion
- What are the parts of an MPLABX project?
=> src\ , prj.X\Makefile and in prj.X\nbproject\configuration.xml, + project.xml
- Steps to compile your FW into a debug'able *.elf?
 1. Get FWrevision from SCM (In: `git checkout revA` ->Out: *.c/.h & configuration.xml)
 2. Update Makefile: `prjMakefilesGenerator.bat` (In: configuration.xml ->Out: nbproject\Makefile*)
 3. `make` to compile sources into *.elf using nbproject\Makefile* (In: Makefile* ->Out: Makefile)
 4. cleanup – eg: remove temp-files,...

End Mod2: compile without IDE

End of Mod2:

Compile without IDE:

Next Mod3a:

MDBCore-based FW-testing with mdbbat